PROJECT DEVELOPMENT PHASE

Business Question and Visualization Report

Date	5 October 2026 (analysis and validation period)	**Team ID**	
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Analysis	**Maximum Marks**	5 Marks

Visualization Development

Visualization development transforms structured data into charts and dashboards that help users compare values, identify trends, and make evidence-based observations. This project uses the global fuel-price dataset in Tableau to explore petrol, diesel and LPG prices alongside Brent crude oil prices, taxation, subsidy levels, region and income level. The Tableau dashboard and Story organize the results for review.

Business Questions and Visualisation

The following questions guide the analysis. Each question is paired with a visualization type and the relevant worksheet or view from the project. Use the published Tableau workbook for the final interactive versions. The report describes observed analytical views; it does not claim causal relationships.

No.	Business question	Visualization / Tableau view	Purpose
1	How do petrol prices vary across countries?	Bar chart: Average Petrol Price by Country	Compare country-level petrol prices and identify higher/lower values.
2	How do global fuel prices change over time?	Line chart: Global Fuel Price Trends	Observe increases, decreases and broad time patterns.
3	How do petrol, diesel and LPG prices compare?	Fuel Type Comparison chart	Compare the three available fuel measures across selected countries or periods.
4	How does tax percentage vary by country?	Bar chart: Tax Percentage by Country	Identify differences in reported tax percentages across countries.
5	How do fuel prices differ by subsidy level?	Subsidy Analysis view	Compare prices across the subsidy categories recorded in the dataset.
6	How do fuel prices differ by income level?	Price by Income Level chart	Compare fuel-price patterns across income-level groups.
7	How do diesel and LPG prices vary by country?	Country-level Diesel and LPG charts	Inspect country differences for the other two fuel types.
8	How has Brent crude oil price changed over time?	Line chart: Brent Crude Oil Trend	Review crude-oil movement over the dataset period.
9	Do fuel prices and Brent crude oil move similarly over time?	Fuel Price and Brent Crude trend comparison	Visually compare trends; correlation or causation is not established by visual comparison alone.
10	How do tax, subsidy, region and income relate to observed fuel-price differences?	Dashboard filters and comparative views	Explore multiple dimensions and check whether patterns differ by group.

Visualization Notes and Interpretation

Project: Global Fuel Prices and Taxation Trends: A Comprehensive Analysis

Dataset overview

The dataset contains 27,468 records across 84 countries, with 10 fields covering dates from 2020 to 2026. Fields include country, region, income level, subsidy level, petrol price, diesel price, LPG price, Brent crude oil price and tax percentage. The available data supports descriptive comparisons and trend exploration.

1. Average petrol price by country

A country-level bar chart makes it easier to rank and compare the recorded average petrol price. Confirm that the aggregation and USD-per-litre unit are clearly stated.

2. Global fuel price trends

A line chart with date on the horizontal axis helps reveal temporal changes. Interpret trends within the available 2020–2026 data window.

3. Fuel type comparison

Use consistent country or date filters when comparing petrol, diesel and LPG so that differences are not caused only by different selections.

4. Tax percentage by country

A sorted bar chart can highlight cross-country variation in the recorded tax percentage. The chart alone cannot explain why a country has a particular rate.

5. Subsidy analysis

Group or filter by the dataset's subsidy-level field and compare price measures. Describe this as an association or observed difference, not proof that subsidies caused the price.

6. Price by income level

Compare the recorded income-level groups using consistent aggregations and clearly labelled axes.

7. Diesel and LPG by country

Country comparisons extend the analysis beyond petrol and help identify whether country rankings differ by fuel type.

8–10. Brent crude, combined dimensions and filters

Review crude-oil and fuel-price lines together, then use dashboard filters to explore region, income and subsidy groups. Note possible time lags, other factors and data limitations.

Dashboard and Story usage

The consolidated Tableau dashboard brings key comparisons into one place. The Tableau Story presents selected views in a guided sequence so a reviewer can follow the analysis from country-level price comparisons to trends and policy-related dimensions. The supporting Flask website provides project information and links to the published outputs.

Evidence and validation checklist

Evidence item	What to verify before submission
Country petrol comparison	Worksheet title, readable country labels, correct average measure and USD/litre unit.
Time-series views	Date axis and time range are correct; no unintended aggregation of dates.
Fuel-type comparison	Petrol, diesel and LPG are labelled clearly and use comparable selections.
Tax and subsidy views	Correct field names, aggregation, and category labels are shown.
Dashboard and Story	All views render; titles, legends, filters and story points are readable.
Published output	Open Tableau Public dashboard and Story links from a browser and confirm they load.

Note: The business questions and view names are tailored to the implemented Tableau analysis. Insert or attach actual screenshots from the Tableau workbook if the evaluator requires screenshot evidence; this PDF does not fabricate screenshots. Team ID is intentionally left blank.

Project period: 5–25 October 2026. Submission deadline: 25 October 2026.